



Evaluation of collision detection and avoidance methods for urban air mobility through simulation

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Received: 24 May 2024 / Revised: 23 October 2024 / Accepted: 28 October 2024 / Published online: 6 December 2024
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Abstract

Urban Air Mobility is a new concept of regional aviation that has been growing in popularity as a solution to the issue of ever-increasing ground traffic. Electric vehicles with vertical take-off and landing capabilities are developed by numerous market companies as a result of the push toward environmentally sustainable aviation. The next stage in this development process would be to define the concept of operation of these conceptual aircraft and then to integrate them with the existing airspace once they are airborne. In addition to coordinating with conventional air traffic and other Urban Air Mobility vehicles, collision avoidance with uncooperative airspace users has to be addressed. Birds and drones of all sizes pose a serious risk to these low-flying aircraft. Innovative collision detection and avoidance techniques need to be employed due to the non-cooperative nature of these airspace users and different performance characteristics of Urban Air Mobility vehicles compared to classical fixed-wing aircraft. The aim of this study is to evaluate the concept of one such system by means of fast-time simulations. This system builds, similarly to the Airborne Collision Avoidance System, on safety envelopes and rule-based collision avoidance to prevent collisions with non-cooperative airspace members. The system is designed to work with all aircraft configurations used for Urban Air Mobility operations. To assess its influence on safety and capacity, different scenarios are modeled by varying parameters, such as intruder type, location, flight path. The parameters assessed are differences in flight time and closest point of approach with and without the collision avoidance system in place. Moreover, the influence of different configurations of Urban Air Mobility aircraft on these parameters is analyzed. The results show that the separation between the ownship and intruder is increased substantially which leads to safe operations at bearable delay costs.

Keywords Air taxi · Birds · Drones · eVTOL · Conflict detection & resolution · Urban air mobility · Air traffic management

1 Introduction

Thanks to the recent urbanization, most people today live in cities [1]. Once nearby villages have developed into sub-urban areas as a result of rapid growth of cities, these areas increase as well. Additionally, there has been a substantial increase in travel time in ground-based transportation systems which is expected to further increase, especially during peak times [2, 3]. To address this issue, numerous mobility

options are being researched. Urban Air Mobility (UAM) is a potential solution that is hypothesized to meet various transportation needs by providing fast and on-demand travel options within and between (sub-)urban areas [4].

However, trying to solve one problem, by adding an additional segment to the already dense airspace, could lead to other challenges. The cruising altitudes of UAM aircraft, referring to air taxis carrying up to four passengers in this paper, are expected between 1,500 and 4,000 ft and cruise velocities ranging from 110 to 322 km/h [5, 6]. These are the altitude bands most abundant with birds and drones [7, 8]. These pose a danger to the safe operation of UAM aircraft due to their non-cooperative characteristics in collision resolution as well as the limited impact resistance requirements of UAM aircraft [9, 10]. An earlier paper by the authors proposed a concept for a UAM Collision Avoidance System (UAM-CAS) for UAM aircraft considering these non-co-operating airspace members (NCOAMs) [11]. This UAM-CAS extends concepts for conventional conflict

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detection and resolution with non-cooperative airspace users such as drones (e.g. [29], [30] by additionally considering birds as potential opponents.

The main element of the concept is an operational decision tree including measures for strategic avoidance prior to flight and measures for tactical as well as emergency avoidance during flight. Similar to the concept of Aircraft Collision Avoidance System (ACAS) of traditional aviation [12], different levels of avoidance maneuvers are performed when an UAM aircraft's safety envelope is penetrated by an intruder. Thereby, three-dimensional safety envelopes for the four most common configurations multicopter, lift to cruise, tilt rotor and vectored thrust were designed. A detailed description can be found in the previous work of the authors [11].

This work focuses on the evaluation of the concept mentioned therein. In the context of the work, 'Evaluation' refers to the process of rigorously analyzing the collision avoidance system for Urban Air Mobility (UAM) vehicles through simulations. The goal is to determine how well the system works in preventing collisions, its impact on flight safety, and its effects on operational efficiency. It is not to be confused with 'Validation', which would typically refer to the process of verifying that the conclusions drawn from a study, or the models and simulations used, accurately represent reality and can be trusted for practical use. It's about ensuring that the system or method being tested not only works as designed but also produces credible and reliable outcomes when applied to real-world scenarios. It was only possible to perform virtual simulations in the scope of this work. Various reasonable assumptions were made in the models used, but an effort was made to be as close to real-life applications.

While it is considered as crucial to introduce operational measures to prevent collision between UAM aircraft and NCOAMs—birds or drones—any solution must also take into account economical aspects [10]. In the United States, the Federal Aviation Administration estimates that flight delays cost airlines 22 billion US Dollars yearly. More than half of the flights in the United States are late (taking account minor delays such as 5 minute or 10 minute delays) [13, 14]. In Europe, a flight delay costs 100 euros per minute on average [15]. This includes everything from extra fuel burn to compensation for passengers. UAM, which is envisioned as a solution to long queues of on-ground traffic, needs to perform within a tight time schedule to prove its efficiency over conventional solutions. The use-case of an inter-city or intra-city air taxi is largely dependent on the time-saving aspect. Therefore, even delays of few minutes can be a make-or-break scenario. Moreover, with the current generation of electric on-board batteries, the maximum time of flight is limited. Therefore, it is crucial, to not only assess the safety impact of any novel concept but also its influence on capacity and punctuality of operations.

The aim of this study is to evaluate the proposed concept of UAM-CAS and to analyze the efficacy of its design. For

this purpose, a sample flight plan of a shuttle service from an airport to a city center represents an UAM use-case [16] and simulated with various conditions in the BlueSky Open Air Traffic Simulator [17]. The simulations included the intrusion of predictable as well as unpredictable non-co-operating airspace member (NCOAM) at different locations along the flight planned route. To evaluate the influence of different safety envelopes, the scenarios were simulated for the four aircraft configurations considered in the original concept.

In the upcoming Sect. 2, the methodology to design and build the simulations is explained in detail. The parameters to be evaluated are presented. In Sect. 3, the results obtained from the simulations for all four configurations are stated and discussed. Finally, in Sect. 4, the results are evaluated and a summary along with the future scope is presented.

2 Methods

In the following Sect. 2.1, details on how the elements of the UAM-CAS are converted into simulation logic is explained. Section 2.2 provides more details about the simulation environment used. Section 2.3 shows the breakdown of the various variables and groups into four topical islands supporting the generalization of the UAM-CAS. In Sect. 2.4, the various phases of the flight and relevant functions which are carried out are introduced. Finally, Sect. 2.5 presents the parameters chosen to evaluate the results from the simulations.

2.1 UAM-CAS

This section provides an overview of the UAM-CAS developed in the authors' previous work [11], to be evaluated within this study to obtain proof of concept. An in-depth description can be found in the original study alongside a visualization of the overall avoidance system as shown in Fig. 1.

The UAM-CAS is based on safety envelopes, similar to the concept of ACAS used in conventional aviation [12]. The safety envelopes are composed of the caution, warning, and collision envelopes as seen in Fig. 2. Whenever an intruder penetrates one of these envelopes, a certain action is initiated. The actions are organized in a decision tree which is divided into strategic, tactical and emergency branches. The strategic phase takes place prior to the flight and as such during the ground phase. In case that NCOAMs are detected close to the take-off area and their movement classified as critical, the departure is delayed. The tactical and emergency phases are active in the airborne phase once the aircraft has initiated its flight. This is also the case when the safety envelopes are activated. An automated tactical maneuver is started when an intruder enters the warning zone of the safety envelope. If the automated maneuver is not successful, the human pilot must take control to avoid the entering

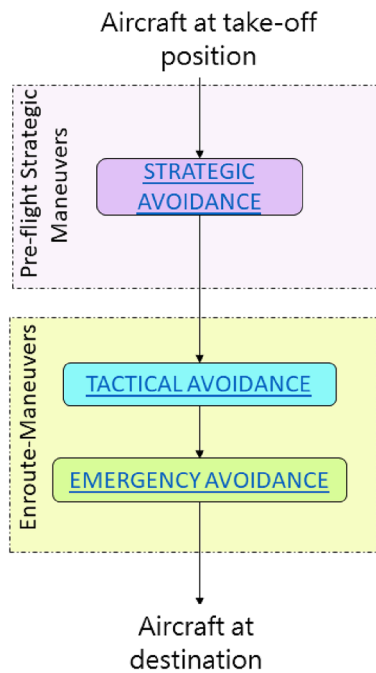


Fig. 1 Decision tree showing the ground and airborne phases of the UAM-CAS, source: [11], used with permission

into the collision zone. This is termed as the emergency maneuver. The size of the safety envelopes strongly depends from the aircraft—the key characteristics to consider are the size as well as the agility of individual configuration. The four configurations most often proposed for the use of UAM operations, namely multicopter, lift to cruise, tilt rotor and vectored thrust are considered in this study.

It is to be noted that in the theoretical study of the concept [11], the authors suggested a different arrangement of the safety envelope zones for forward and vertical flight: Intersecting circles meeting at a common point for forward flight and concentric circles for vertical flight. In contrast, the zones are organized as concentric circles for all phases of flight to facilitate the simulations within this work. The arrangement of concentric circles throughout the flight can lead to a slightly increased flight time.

2.2 BlueSky Open Air Traffic Simulator

To evaluate the presented concept of an UAM-CAS, its functionality was simulated in a fast-time simulator. For this purpose, the BlueSky Open Source Air Traffic simulator developed by Delft University of Technology was selected. BlueSky is an open source simulator used for research in

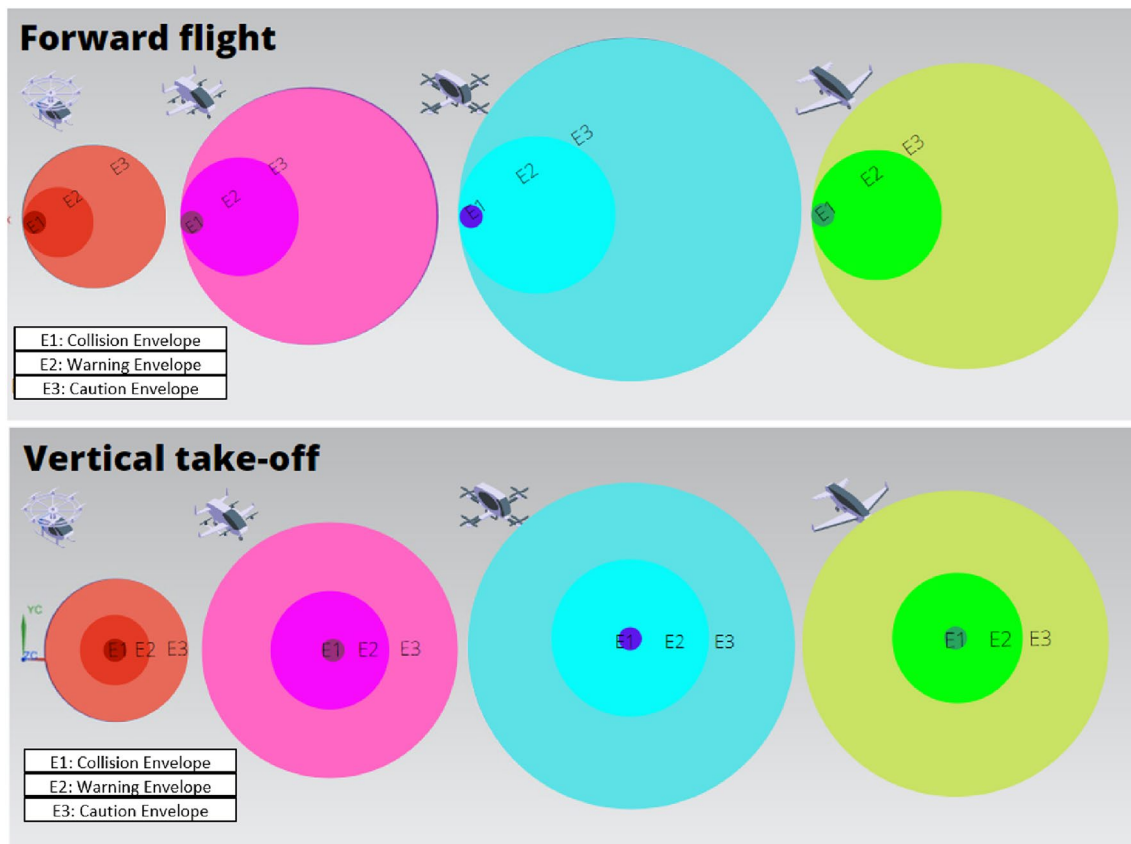


Fig. 2 Collision, warning and caution envelopes for various configurations in forward and vertical flight, source: [11], used with permission

domains of air traffic management and to optimize air traffic flow [17]. Its open source character as well as the option to add functionalities in independent plugins made it suitable as a platform for this study. Relying on existing plugins for the representation of bird movement as well as collision detection in the simulator [18], multiple plugins were developed within this work to represent the UAM-CAS. These included plugins for collision avoidance, the visualization of aircraft and intruders with their flight paths as well as the definition of the safety envelopes depending on the UAM aircraft configuration.

During runtime, the simulation itself is operated from the main simulation window of BlueSky. An additional window provides the user with the current state of the UAM-CAS, currently active automated avoidance maneuvers or, in case of an emergency maneuver, the request for execution by the operator. During the simulation, all relevant parameters were logged and stored for analysis.

2.3 Variables of UAM-CAS: Four Islands

The simulation tasks were divided into four topical sub-tasks called ISLANDS. Once defined, the individual components covered by each island can be swapped easily in case the same analysis needs to be performed for any other location or flight characteristics. Figure 3 shows the islands and their relations [11].

Island 1 is related to the flight path. Island 2 contains all characteristics of the ownship which need to be considered. Island 3 deals with intruder and their types. Island 4 contains the definition of the collision avoidance maneuvers. Out of the four islands, three are standalone and can be populated with content without any dependencies on the other islands. The last island 4 covering the maneuvers overlaps with the other three islands.

- 1) ISLAND 1- Geography and Flight Path: In this island, all parameters related to the geography are taken into consideration. This includes, among others, the position of the take-off and the landing vertiports, preferred and alternate routes to be flown and alternate landing vertiports.
- 2) ISLAND 2- Aircraft Type: This island deals with the ownship aircraft. The configuration to be considered must be clearly defined. For this study, the four configurations like the multicopter, lift to cruise, tilt rotor and vectored thrust were modeled based on the pre-existing open-source kinematic aircraft performance model in BlueSky [19, 20] as well as information from Electric Vertical Take Off and Landing (eVTOL) manufacturers [21–24], as described in detail in [11]. Only the motion of the ownship through space is considered without reference to any study of external forces. Based on the configuration, the dimensions of the safety envelope also

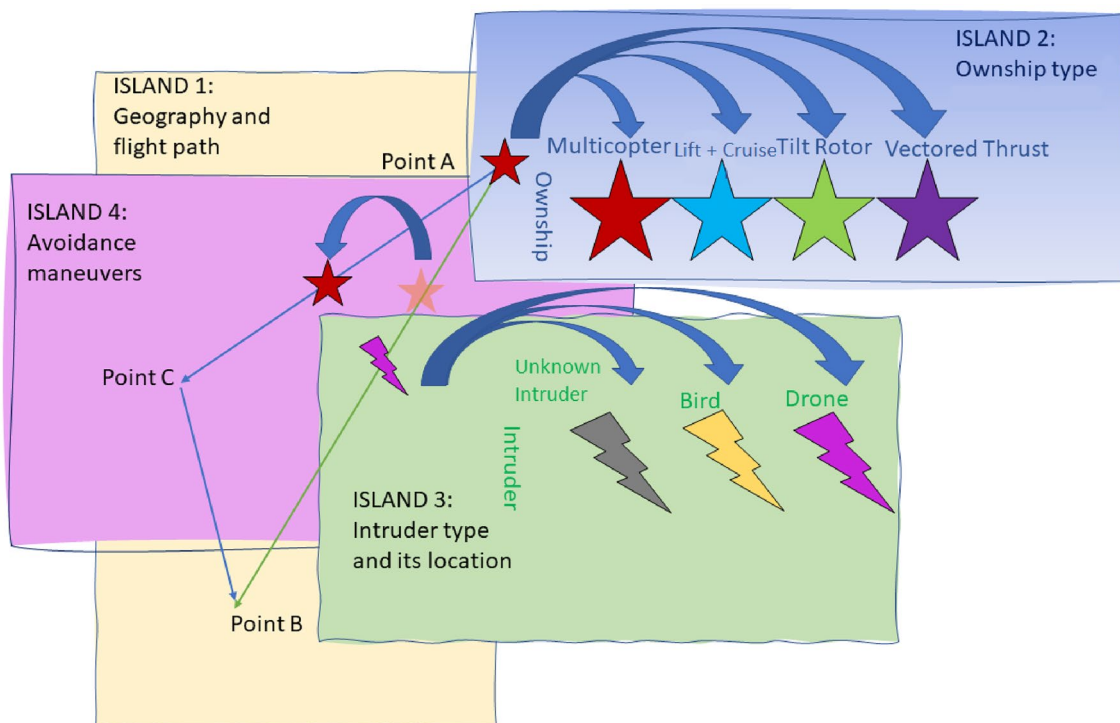


Fig. 3 UAM-CAS tasks grouped in four islands

differ. Some elements of the tactical collision avoidance logic are dependent on the information of the configuration of the ownship. These are contained in that island as well.

- 3) ISLAND 3- Intruder Type and its Location: This island deals with the intruder and related factors like its type, trajectory and location of occurrence. All intruders are NCOAMs. The NCOAMs can display predictable behavior or can be unpredictable. Drone obeying Right of Way rules of the sky or birds flying away from the ownship due to noise or size is considered predictable behavior [25]. The intruder namely a drone or a bird is programmed to appear at various locations head-on, from the left or right or on the same path as the ownship. These directions are considered so that they can be linked to the conventional Right of Way rules of air. The trajectory of the intruder is pre-defined for repeatability. Birds were assumed to be appearing on the route from the vicinity of Isar River, where they might be nesting owing to their conservation of their natural habitat. Drones appeared in the vicinity of the Allianz arena, where they might be used for photography.
- 4) ISLAND 4- Avoidance Maneuvers: This island defines the collision avoidance maneuvers which are required to avoid encounters with NCOAMs. The incorporation of the required maneuvers within BlueSky is described in section 2.4

2.4 Simulation Set-up and Scenarios

After implementing all information pertaining to the four islands in BlueSky, the scenarios are defined. The selected flight plan represents a shuttle service from the airport of Munich (EDDM V1) to the main railway station of Munich (MUC HBF V2). For an alternate landing site, various locations, such as the top of high-rise buildings, helipads of hospitals, and big community grounds, are possible. For the demonstration in this study, the helipad on the Oberschleissheim airport (EDNX) was selected as the dedicated alternate vertiport V3 (EDNX V3). Figure 4 shows an overview of the flight routes.

The general scenario, which is performed for each configuration, is set up as follows: the UAM-CAS starts with a pre-flight strategic maneuver for an UAM aircraft at V1(EDDM). Before take-off, the pilot scans for intruders using available systems like avian Radio Detection And Ranging (RADAR) or Notice to Airmen (NOTAM). If intruder activity is detected on the regular route 1 and active for over five minutes, the scanning process is repeated for the alternate route 2 via V3. If that intruder activity pertains for over five minutes on route 2, the take-off is postponed for later consideration. Although longer, this approach is a preferable compromise to entirely canceling the flight.

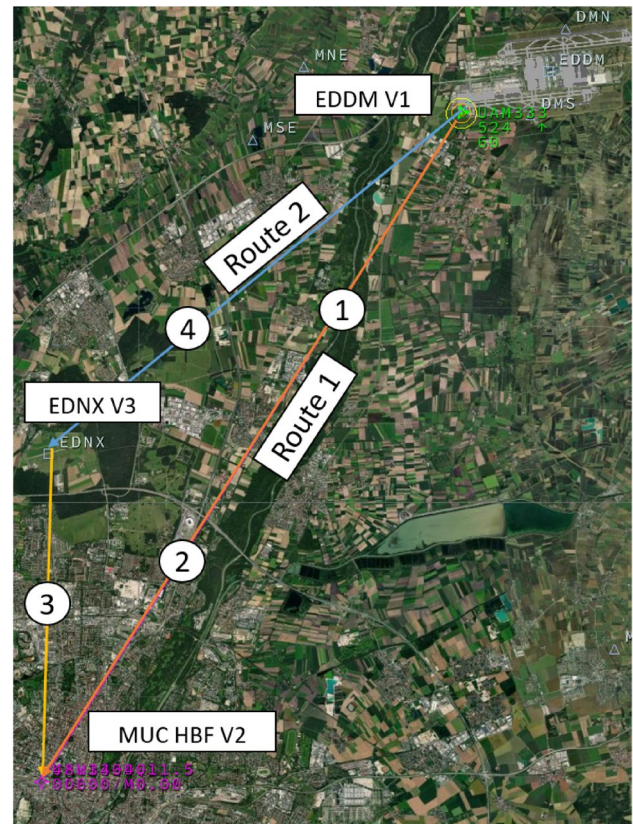


Fig. 4 Vertiports, ownship with safety envelopes, available routes and location of intruders in scenarios 1–14 on the map view in BlueSky

After take-off, the aircraft climbs to 1000 ft (304.8 m) above ground level (AGL). For the vertical phases, the climb rate of 1.7 m/s is applied for all configurations [21–24]. Once the ownship reaches 1,000 ft, the forward flight phase of the mission is initiated. For the forward flight, cruise speed of 78 m/s is applied. The flight is continued until an intruder appears which triggers the tactical or even emergency phase (Table 1).

According to the decision tree, the strategic move is executed on the ground and prior to flight. There are different steps that need to be taken in the tree's strategic maneuver branch, depending on whether and where birds are present (see Fig. 5). It is assumed that a UAM aircraft with a vectored thrust configuration is waiting at EDDM V1 for take-off. The pilot scans the surroundings to ensure an obstacle-free take-off path. It is anticipated that the pilot will have access to NOTAM or, if available, real-time sensing systems like avian RADAR, to obtain information about the invaders' whereabouts. After analyzing this data, the pilot must wait five minutes in case that intruder activity was observed or reported on route 1. If there are no trespassers, the pilot can initiate the departure on time. The pilot may contemplate taking off and flying route 2 if the intruder activity on route 1 does not stop after five minutes. However, route 2 will also

Table 1 Theoretical flight values of flight parameters of the vectored thrust ownship

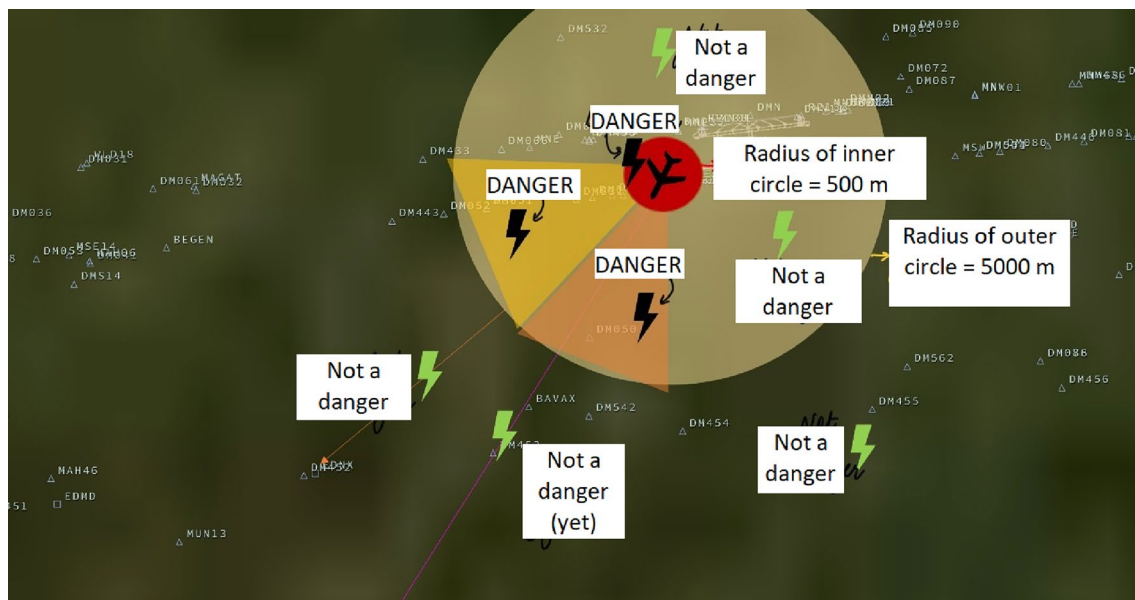
	Horizontal cruise	Vertical climb	Vertical descent	Theoretical value for total flight
Distance on route 1: d_{R1} (m)	26000	304.8	304.8	26610
Distance on route 2: d_{R2} (m)	30000	304.8	304.8	30610
Ground speed of vectored thrust ownship: S_{VT} (m/s)	78	1.7	1.7	
Time for route 1: $t_{R1} = d_{R1}/S_{VT}(s)$	333.33	179.29	179.29	692
Time for route 2: $t_{R2} = d_{R2}/S_{VT}(s)$	384.61	179.29	179.29	744

undergo a similar check for intruder activity before to take-off. The pilot will have to wait an additional five minutes if it turns out that route 2 is the subject of intruder activity. The pilot can take off using the alternate path if there are no trespassers. After five minutes, the pilot must check route 2 again for signs of intruder activity. If the trespassers are still at the take-off location, the decision would be made to delay take-off and think about taking off later. This comprises the **Delay on Ground** which were listed in Table 3.

Once the ownship is airborne, it flies toward V2. To evaluate the collision avoidance methods, NCOAMs which vary in type (drone or bird), their predictability and direction of approach would appear at a pre-defined time and pre-defined co-ordinates (locations 1–4 in Fig. 4). In every scenario, one ownship encounters one NCOAM. Once the NCOAM penetrates the warning envelope, the automated tactical maneuver begins. The ownship hovers or changes its direction of flight (based on its configuration). If this does not solve the problem, the human pilot has to intervene to avoid the collision. This is termed as the emergency maneuver. Once safe separation is ensured between the NCOAM and ownship,

the mission is continued and the ownship will continue flying toward V2. The tactical and emergency maneuvers may cause disruptions in the original flight plan and can lead to increased flight time. This raise is used as indicator of the efficacy of UAM-CAS and is thus considered as **Delay in Air**. For an overall evaluation, the ground and airborne delays were added to get the **Total Delay of Flight** (Table 4).

When the ownship is at V1 and ready to depart, one of the strategic maneuver's five branches must be true for take-off. For instance, as seen in Fig. 6, in the case of a vectored thrust ownship, it's assumed no intruder activity is near the airport, making branch 1 true. If the ownship encounters intruders after take-off, it refers to the tactical maneuver segment for collision avoidance, offering seven options. Assuming the ownship faces a head-on conflict with a drone on route 1 and identifies it, the ownship, being of vectored thrust configuration, will turn right as dictated by branch 7 of the tactical segment. The conflict may resolve, or an emergency maneuver might be needed, with two possible outcomes in this segment. If there are no substantial issues, the pilot continues to V2. Summarizing,

**Fig. 5** The various locations of the intruder with respect to the ownship along with indication on whether they pose a danger

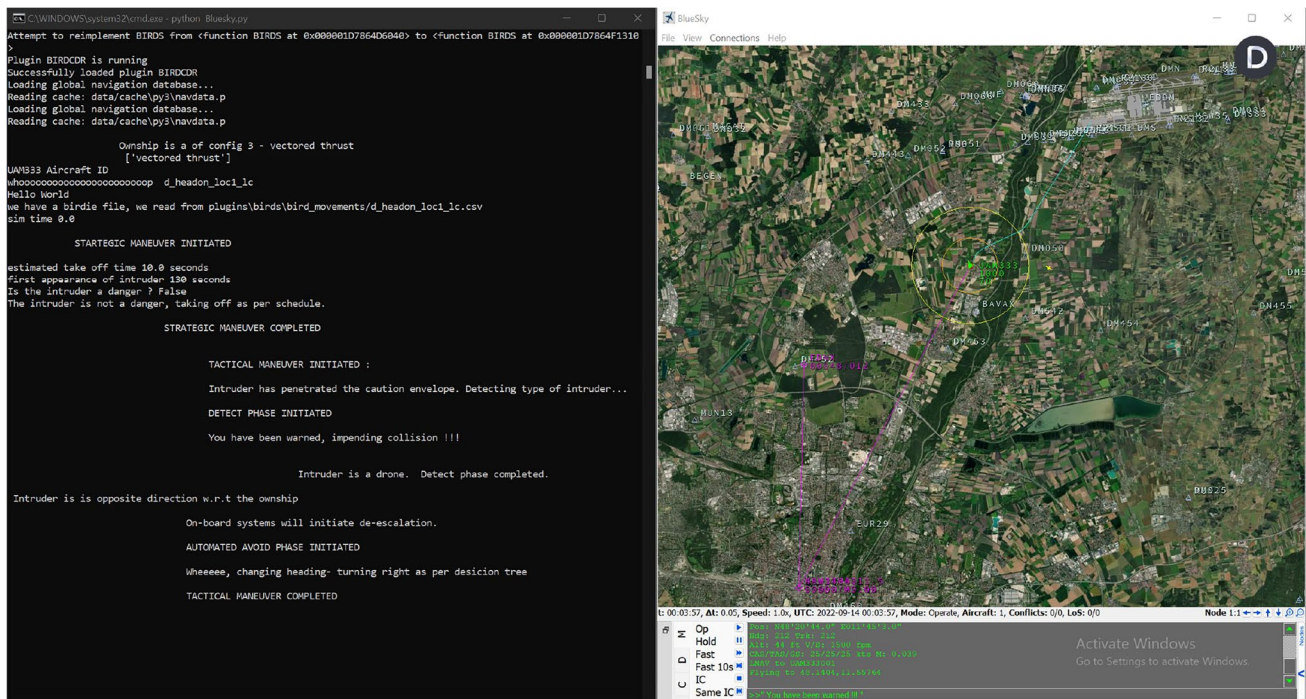


Fig. 6 Simulation screen view with Anaconda command prompt on the left and map view of BlueSky on the right of scenario 5

one true branch from each maneuver segment is required, leading to a total of 70 possible combinations (5 strategic options * 7 tactical options * 2 emergency options). This study simulates five scenarios for strategic maneuvers, seven for tactical, and two for emergency, considering 14 cases in total. The total flight time is calculated by adding the times from these individual segments.

In the first 13 scenarios, the automated systems along with the human pilot act to resolve the conflict as per the steps mentioned in the UAM-CAS logic. In scenario 14, the conflict is not resolved by tactical and emergency maneuvers and the intruder collides with the UAM aircraft. This scenario is included to investigate the consequences of a failure of the UAM-CAS. Table 2 provides an overview of all scenarios as well as a detailed description of the airborne-phase maneuvers. The scenarios are simulated for all 4 configurations.

UAM-CAS takes into consideration the individual flight physics and handling quality depending on individual aircraft configurations, particularly in terms of collision avoidance. Each configuration has its strengths and weaknesses, impacting collision avoidance capabilities and efficiency in different flight phases. For this study, the four configurations currently dominating the market are considered [21–24]:

- **Multicopter:** This configuration is characterized by multiple propellers in a single plane. This configuration is most efficient for vertical operations but less so for cruise flight due to the absence of wings.
- **Lift to Cruise:** This configuration merges multicopter capabilities with those of standard aircraft for vertical take-off and cruise flight, offering a compromise between efficiency and noise emission.
- **Tilt Rotor:** This configuration utilizes slowly rotating multi-blade propellers to mitigate disadvantages, but faces challenges in technological complexity and transition flight dynamics.
- **Vectored Thrust:** Similar to tilt rotors but with enclosed motors forming a jet, offering advantages in long-range flights but requiring significant energy for hovering.

2.5 Test parameters to be evaluated

The 14 scenarios which were implemented in BlueSky, are a set of different situations that can happen during the flight of a UAM airport shuttle. To analyze the efficient operation of the UAM-CAS, the following parameters were evaluated in this study.

Closest point of approach To evaluate the efficacy of the UAM-CAS, one of the parameters to be analyzed from the series of simulations is the Closest Point of Approach (CPA) of the intruder with respect to the ownship. This is the shortest distance between the ownship and the intrusion observed

Table 2 Description of airborne phase maneuvers for each intruder scenario

Intruder scenario					Automated avoidance maneuver once intruder enters caution envelope	Emergency maneuver once intruder enters warning envelope
Sce- nario num- ber	Type	Intent / Behavior	Location of con- flict	Direction		
1	Drone	predictable	1	right	Hover at position, Forward velocity = 0, Drone has RoW	
2	Drone	unpredictable	1	right	Hover at position, Forward velocity = 0, Drone has RoW	Turn left
3	Drone	predictable	1	left	Continue flight, Ownship has RoW	
4	Drone	unpredictable	1	left	Continue flight, Ownship has RoW	Turn right toward V3- EDNX
5	Drone	predictable	1	head-on	Ownship of configuration - 4, turn right	
6	Drone	unpredictable	1	head-on	Ownship of configuration-4, turn right	Continue going right toward V3- EDNX
7	Drone	predictable	1	same direction	If ahead of ownship, drone has RoW, change path	
8	Drone	unpredictable	1	same direction	If ahead of ownship, drone has RoW, change path	Increase separation and continue on parallel path
9	Bird	predictable	1	head-on	Hover and descent to 800 ft, Forward velocity = 0	
10	Bird	unpredictable	1	head-on	Hover and descent to 800 ft, Forward velocity = 0	Turn right toward V3-EDNX
11	Drone	unpredictable	2	head-on	Ownship of configuration - 4, turn right	Continue going toward V2- MUC HBF
12	Drone	unpredictable	3	head-on	Ownship of configuration - 4, turn right	Continue going toward V2- MUC HBF
13	Drone	unpredictable	4	head-on	Ownship of configuration - 4, turn right	Continue going toward V3- EDNX
14	Bird	unpredictable	1	head-on	Hover and descent to 800 ft, Forward velocity = 0	Turn right toward V3-EDNX

throughout the simulation. The scenarios were set up to result in a collision (i.e., CPA = 0 m) without any intervention by UAM-CAS. Hence, the recorded increase in CPA can be interpreted as the safety measure.

Time of simulated flight $T_S(S)$ and Delay in Air $D_A(S)$
Based on the distance and performance data of the aircraft, the theoretical total time of flight is calculated. For each scenario, a mission is simulated in BlueSky without any disturbance from the intruders. This is done to evaluate the performance model of the ownship used for the simulations. As is to be expected, the total time of this simulated undisturbed flight is exactly same as the theoretical total time of flight, the theoretical total time of flight serves as a benchmark to evaluate each of the scenarios requiring collision avoidance due to intruders.

3 Results and discussion

To evaluate the utility of the proposed UAM-CAS system, two parameters were evaluated namely CPA and total time of simulated flight $T_S(S)$.

As described, the delays were calculated for the ground (strategic) and airborne (tactical and emergency) phase separately before combining them to obtain the values for the total delay.

3.1 Ground phase

3.1.1 Delay on ground: $D_S(S)$

The ground phase consists of the amount of time the aircraft has spent on the ground at V1, ready for departure. The

Table 3 Outcome and causes in strategic avoidance in ground phase

Possible outcomes	Cause	Delay (sec)
Scheduled Departure on Route 1	NCOAM headed to Route 1 or in inner circle	0
Departure after delay of 5 min on Route 1	NCOAM stays in vicinity of Route 1	300
Departure after delay of 6 min on Route 2	NCOAM headed to Route 2	360
Departure after delay of 11 min on Route 2	NCAOM stays in the vicinity of Route 2	660
Flight is postponed	NCOAM activity has not cleared	infinite

delay in duration between ready for departure and effective departure is caused by intruder activity in the vicinity of the vertiport and is independent of the aircraft configuration. As shown in Table 3, there are five different outcomes possible as discussed in the previous work of the authors [11]. The best case to fly route-1 is at the scheduled departure time. The worst case to fly route-1 is with a delay of 300 s (or 5 min). In case that the aircraft has to switch to route 2 due to intruder activity on route 1, take-off takes place the earliest after 360 s (6 min). If there is intruder activity on route 2 as well, then the take-off is delayed by 660 s (11 min) which is represented by scenario 12 and 13. In the worst case, the take-off is postponed to a later time if there is continued presence of intruders even on route 2 as well. As it can be seen in Table 2, the ownship flies on route 1 toward V2 in scenarios 1–11 and 14. Therefore, it is implied that in scenarios, the worst case ground delay of 300 s would be considered and in scenarios 12–13, a worst case ground delay of 660 s would be considered. As the ownship is not yet moving, the parameter of CPA is not relevant. The delay calculated in Table 3 needs to be added to the delay of the airborne phase to evaluate the total flight delay.

3.2 Airborne phase

3.2.1 Closest point of approach (CPA)

In Fig. 7, the CPA values for scenarios with UAM-CAS and without UAM-CAS are compared for a vectored thrust ownship to analyze the safety effect of the UAM-CAS. It can be observed that except in scenario 10, in all other scenarios, there is a substantial increase in CPA when the UAM-CAS is active. This drop can be as high as 98 %. This indicates the value of an UAM-CAS to maintain a safe separation between aircraft NCOAMs. Without such a system, an intruder can come dangerously close to the ownship, causing a near miss if not a collision.

In Scenario 10, a bird was encountered head-on en-route V2. The automated tactical maneuver was to hover and descent to 800 ft, with zero forward velocity. Although hovering is an energy-intensive maneuver, it was chosen due to the unpredictable behavior of the bird. In this scenario, the tactical maneuver did not resolve the conflict

but made it even worse. The ownship got too close to the intruder which resulted in the CPA being low. The human operator took over and performed the emergency maneuver and brought the ownship to safety (Fig. 8).

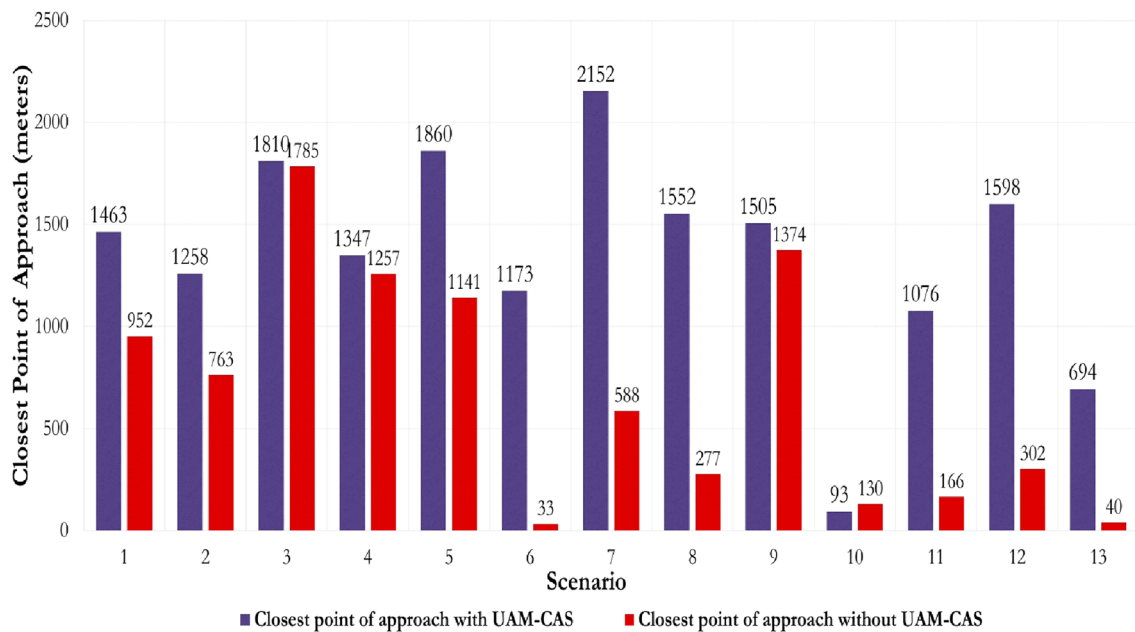
After the in-depth analysis of the concept, the influence of the configuration is studied. It has been assumed that multicopter and lift to cruise configurations would use hover as a means of collision avoidance, whereas the tilt rotor and vectored thrust would turn (mostly right) to get out of harm's way. Multicopter and lift to cruise configurations have major thrust-generating components in the vertical axis, whereas tilt rotor and vectored thrust have thrust production mainly in the horizontal axis. This gives the respective aircraft more control in the thrust axis and therefore the collision avoidance is chosen in this manner. When looking at Fig. 9, it can be observed that, CPA has indeed increased for all four types of ownships. As defined in [11], multicopter and lift to cruise ownships descended to avoid a head-on collision with drones, whereas ownships of tilt rotor and vectored thrust type turned right. Both of these techniques resulted in successfully evading the intruder. The tilt rotor boasts the most extensive 3D envelopes, closely trailed by the vectored thrust configuration. However, there's a noticeable contrast in envelope dimensions between the multicopter and tilt rotor. The caution envelope of the multicopter, which is the biggest of the three envelopes, is even smaller than the warning envelopes of the other three configurations. This would imply that if a uniform envelope dimension were adopted for all the four configurations, than the multicopter would have to change direction of flight, velocity or altitude a lot earlier before the danger of the intruder was imminent, leading to inefficient flight performance. Unlike commercial aviation, where aircraft within the same category typically shares similar configurations, the UAM landscape presents a varied scenario. This underscores the necessity of adopting a configuration-specific approach for UAM.

3.2.2 Simulated flight time $T_S(S)$ and Delay in Air $D_A(S)$

From Fig. 10 for the vectored thrust ownship, it can be observed that the total time of simulated flight for most scenarios (except scenario 1 and 2) is nearly equal to the total time of theoretical flight with no intruders present

Table 4 Values of total delay for entire mission of different scenarios

Intruder scenario					Delay Information		
Scenario number	Type	Intent / behavior	Location of conflict	Direction	Max delay in ground phase: $D_G(s)$	Delay in airborne phase: $D_A(s)$	Total delay during entire mission: $D_T(s) = D_G + D_A$
1	Drone	predictable	1	right	300	531	831
2	Drone	unpredictable	1	right	300	513	813
3	Drone	predictable	1	left	300	1	301
4	Drone	unpredictable	1	left	300	3	303
5	Drone	predictable	1	head-on	300	29	329
6	Drone	unpredictable	1	head-on	300	62	362
7	Drone	predictable	1	same direction	300	64	364
8	Drone	unpredictable	1	same direction	300	60	360
9	Bird	predictable	1	head-on	300	32	332
10	Bird	unpredictable	1	head-on	300	38	338
11	Drone	unpredictable	2	head-on	300	3	303
12	Drone	unpredictable	3	head-on	660	42	702
13	Drone	unpredictable	4	head-on	660	11	671
14	Bird	unpredictable	1	head-on	300	105	405

**Fig. 7** Closest point of approach with UAM-CAS and without UAM-CAS for each intruder scenario

(maximum difference of 93 s) Therefore, by adapting the proposed UAM-CAS, the ownship can maintain safe separation from intruders without substantially prolonging the total flight time. As seen in Fig. 11, in scenario 1 and 2, the delay in flight is quite high, with a maximum difference of 531 s. This is due to the fact that the intruder, in this case, a drone entering the safety envelope from the right with respect to the motion of the ownship, lingers in the

vicinity of the ownship for a longer time. The intruder, in this case a drone, coming from the right has Right of Way and therefore the tactical avoidance maneuver of a UAM aircraft (regardless of the configuration), to avoid collision with this type of an intruder is to hover in place. As long as the intruder is in the vicinity of the ownship, it will continue hovering. The time of hover might be a critical factor for eVTOL aircraft as the energy required to hover

Fig. 8 Comparison of CPA for predictable and unpredictable intruders

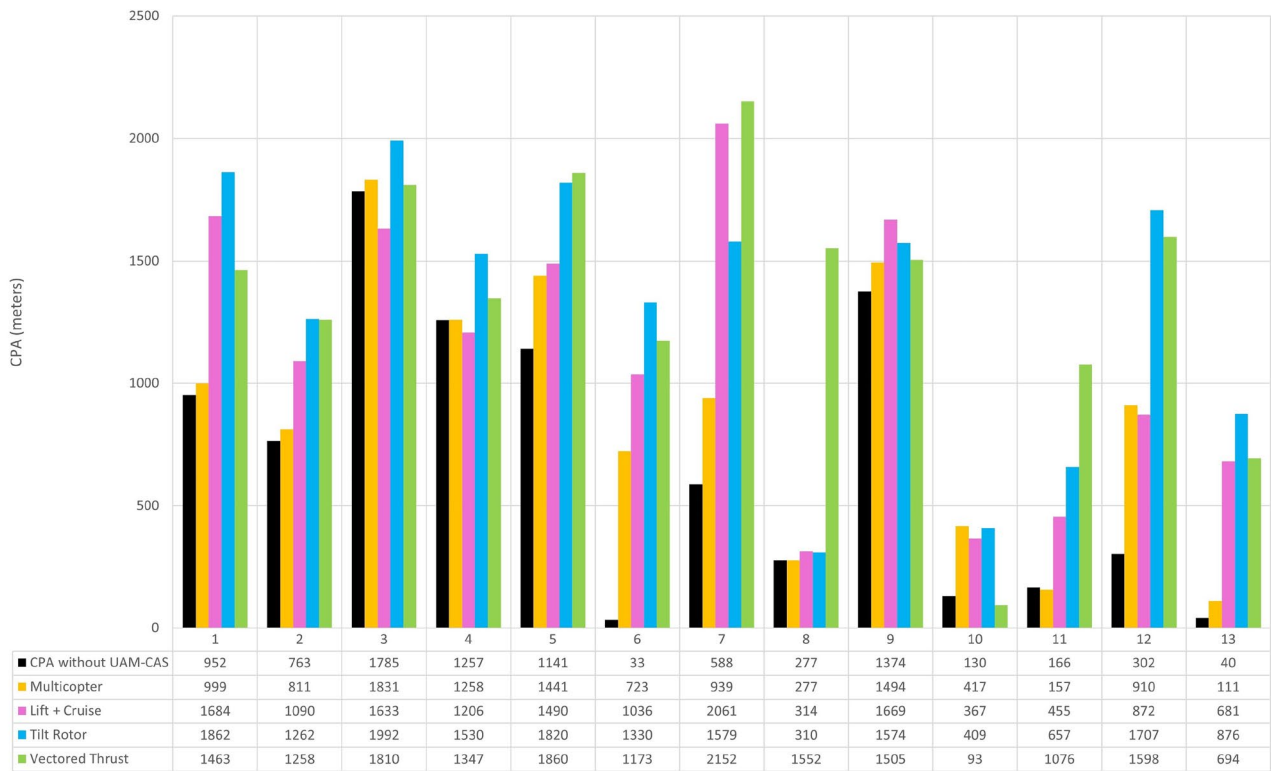
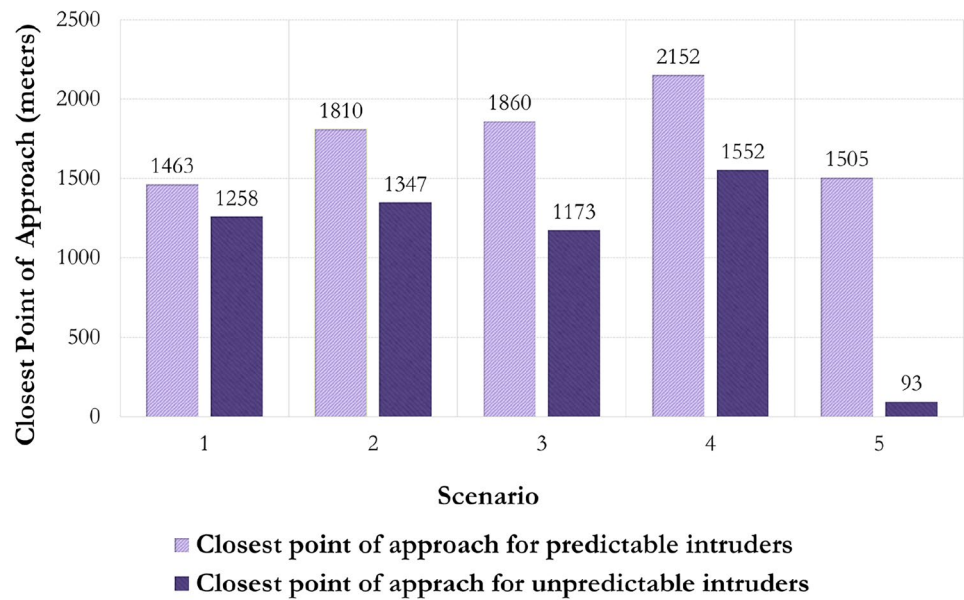


Fig. 9 Comparison of CPA for all four types of ownships. The black bar shows the value of CPA for the vectored thrust ownship without an active UAM-CAS

is more than the energy required for forward flight [26]. Therefore, the avoidance behavior needs to be optimized. Moreover, as seen in Fig. 12, the total time of simulated flight is almost similar or slightly more (maximum difference of 33 s) for scenarios with unpredictable intruders as

compared to scenarios with predictable intruders. Highlighting the influence of configuration, the delay in flight for the multicopter and lift to cruise configurations is more when compared to the other two configurations. This may be due to the fact that multicopter and lift to cruise use

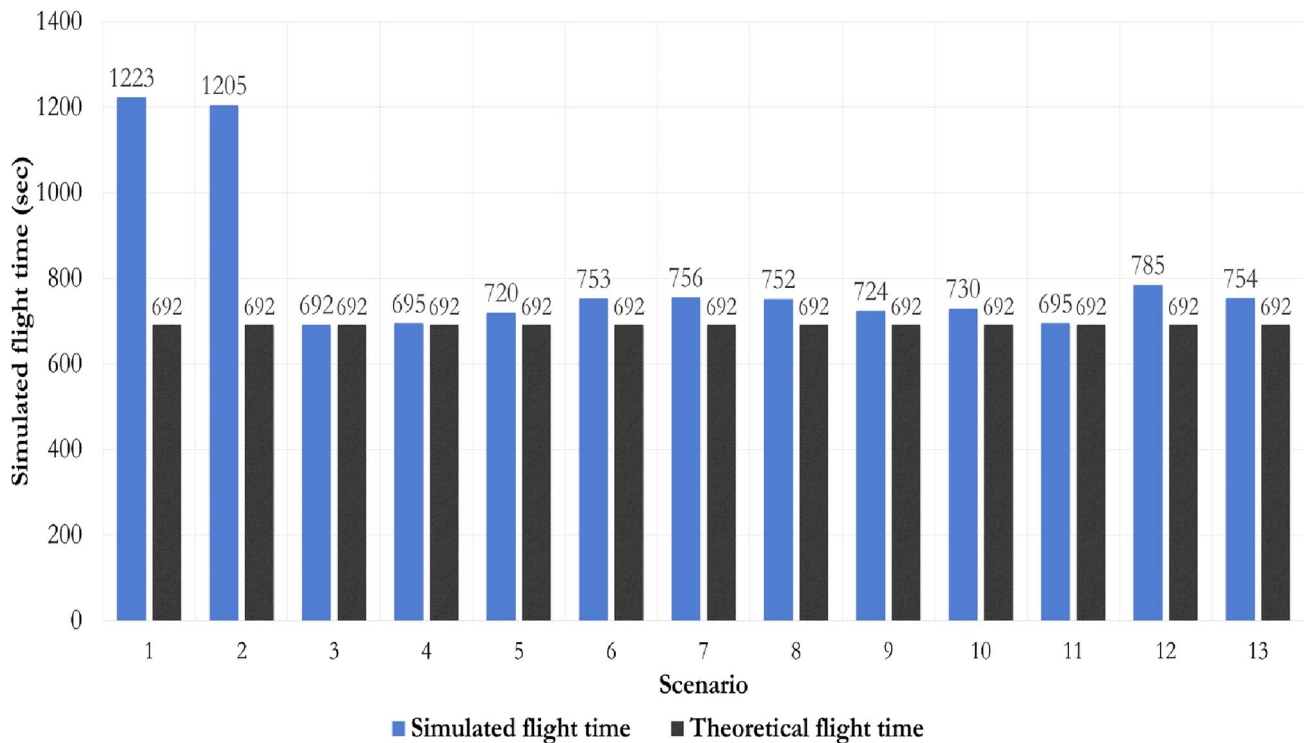


Fig. 10 Total time of simulated flight for each intruder scenario

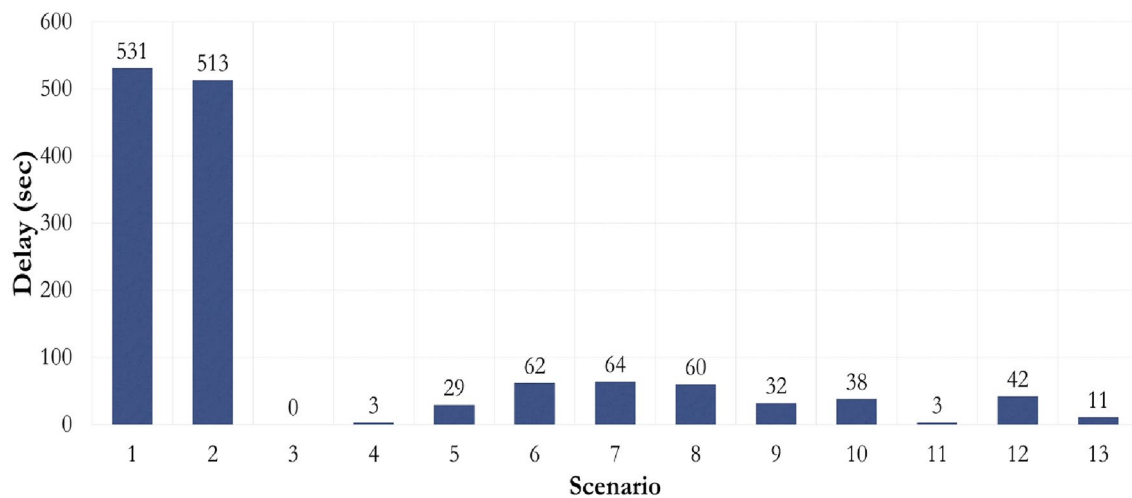


Fig. 11 Delay for each intruder scenario in the airborne phase. Note: Detailed description of the scenarios can be found in Table 2

combination of hover and descend maneuvers to resolve the conflict which is evident from Fig. 10.

If the delay is compared for all ownship types, as in Fig. 13, it can be observed that the delay is the highest for multicopter followed by lift to cruise ones. Ownships of multicopter and lift to cruise configurations transition into

hover in case of the threat of head-on collision with a drone. Therefore, the aforementioned increase in the delay could be due to long hover times. The delay is the least for the tilt-rotor configuration of the ownship. To avoid the intruder, the tilt rotor descends and this could be an influencing factor in the reduced delay.

Fig. 12 Comparison of total time of simulated flight for predictable and unpredictable intruders

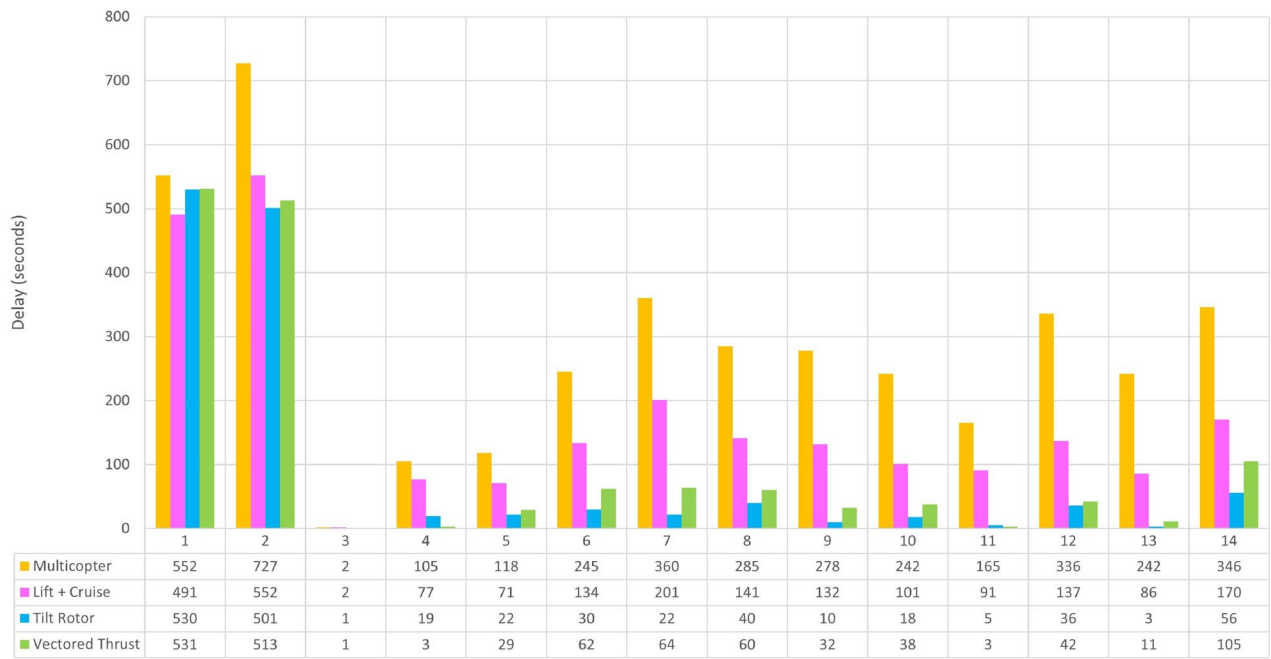
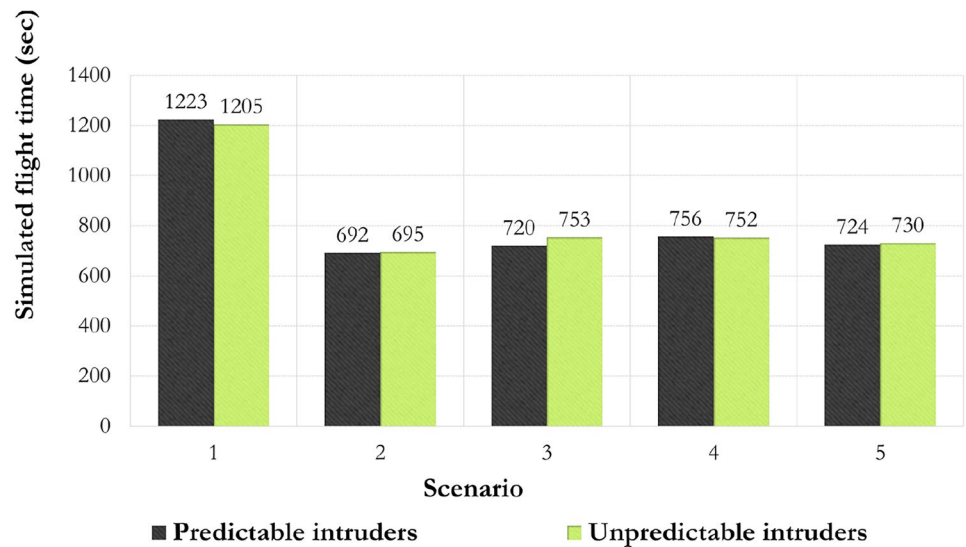


Fig. 13 Comparison of delay in the airborne phase for all four types of ownships. Note: Detailed description of the scenarios can be found in Table 2

3.2.3 Total delay for the entire mission $D_T(S)$

The delay components of the ground phase and the airborne phase were summed up as shown in Table 4 to find the total delay for the entire mission for the vectored thrust ownship. In scenarios 1–11, the ownship flew on route 1, and as shown in Fig. 3, the worst-case delay for that route is five minutes. Therefore, $D_G(S)$ is 300 s for scenarios 1–11 and 14. The ground phase delay is now added to the delay of the

airborne phase $D_A(s)$. For example, in scenario 9, the ownship faced a delay of 300 s on the ground and then another 32 s to evade the head-on bird during flight, resulting in total delay for the entire mission to be 332 s. The maximum total delay $D_T(S)$ for all scenarios is obtained as shown in Table 4. In a similar manner, for route 2, the worst-case ground phase delay $D_G(S)$ is 660 s which corresponds to scenario 12 and 13. The values are visualized in Fig. 14. It can be observed that the delay of the ground phase is the largest component

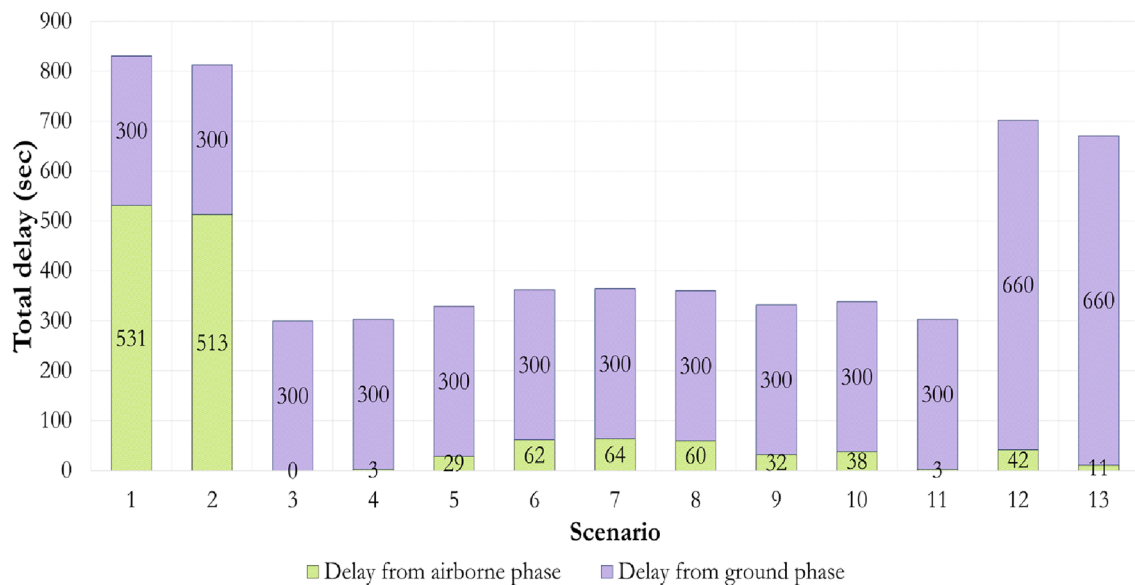


Fig. 14 Maximum total delay comprising of the ground phase component as well as the airborne phase component for each scenario for the entire mission

and substantially delays the airport shuttle operations when compared to the airborne delay. Hence, emphasis should be put on measures to minimize ground delays.

4 Conclusions

Within this study, a previously proposed UAM-CAS concept was evaluated and its efficacy was analyzed. Therefore, the parameters of CPA and delay were assessed for the four most common UAM aircraft configuration. From Sect. 3, it can be concluded that the ground phase is responsible for larger delays than the airborne phase. This could be counteracted by reducing the take-off delay duration of five minutes in case of critical NCOAM movement. In addition, patterns and time windows during which intruders are the most active near V1 should be studied and strategies be developed based on the situation present at the intended departure time. Systems like avian RADAR could facilitate the required detection and monitoring of birds [27, 28]. A prerequisite is a solid ground infrastructure with sufficient capacity for the harassment of wildlife and solutions to protect the area from NCOAMs.

On the other hand, with an average delay of 107 s, the airborne delays are mostly negligible, even at this early concept state. However, solutions to shorten the delays due to unpredictable intruders need to be found. This is especially true for the multicopter and lift to cruise configurations which hover in case of observed intruders. In addition to requiring a lot of power, the delays also quickly accumulate. Hence, alternative avoidance maneuvers should be developed to

counter the ground delays and analyzed with regard to their impact on safety and efficiency. For example, a workaround in these scenarios could be to re-evaluate the situation after a threshold time and avoid the conflict in air through change in heading or altitude. Such alternative maneuvers should be assessed in the context of the maneuvering capabilities of the different configurations and evaluated again through simulation. For the other scenarios and configurations, the delays are reasonable and UAM aircraft can be operated with the current generation of batteries when UAM-CAS is active. Moreover, the proposed maneuvers themselves can be successfully carried out. From Fig. 9, it can be concluded that using UAM-CAS, the CPA can be reduced substantially by an average of 72 %.

For an UAM-CAS like the one proposed here to be implemented, several prerequisites need to be fulfilled. First, sensors on the ground or in the air to reliably detect and identify NCOAMs are required. Second, to predict their movement in the near future to decide about the necessity of avoidance maneuvers, models need to be developed and thoroughly tested. Lastly, these elements, alongside the suggested refinements to the UAM-CAS should be evaluated for various conditions, such as aircraft configurations, terrain characteristics or weather. Additionally, factors like detection of false positives and true negatives, as well as delayed reporting when conflicts cannot be resolved immediately, must be taken into account. The predictability was also considered without any difficulties. It was assumed that the on-board sensors would be precise and functional throughout the mission. No uncertainties were taken into this work. Ensuring

satisfactory sensor reliability in reality would be paramount in future work.

The avoidance maneuvers could be further refined based on latest aircraft performance data. By performing further tests with other UAM configurations and with a bigger sample size of scenarios, more reliable results can be achieved. UAM-CAS could act as a basis for a novel Collision Detect and Resolution (CDR) method for eVTOL flight. Based on the evaluation of technological advancement, there is potential to explore the requirements for autonomous CDR. However, this exploration must be substantiated through rigorous testing, given the inherent high level of risk involved. Furthermore, it's conceivable that future iterations of existing UAM aircraft may exhibit improved performance characteristics, necessitating a reevaluation of previously presented calculations.

In the hypothetical flight scenario, intruders were addressed singularly and only once per flight. For instance, the consideration of a flock of birds as an intruder is one such example. In this study, the discussion regarding birds focuses on individual instances. Further simulations could explore additional combinations and frequencies, evaluating actions outlined in the UAM-CAS logic. Refinement of the 3D safety envelope is also feasible, allowing for more precise shapes such as tear drop-like configurations instead of circles to take into consideration the effect of relative motion during forward flight. This study proposes a pathway for the future integration of UAM, offering a potential avenue for realization in the near term

Author contributions I.P. set up the conceptual framework of the paper and performed the simulations. She led the writing of the manuscript. S.F.A. contributed to the study design as well as to the writing and revising of the manuscript. I.C.M. contributed to the setting up of the simulations, the study design as well as to the writing and revising of the manuscript.

Funding Open Access funding enabled and organized by Projekt DEAL.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. The authors have no conflict of interest to declare that are relevant to the content of this article.

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